Common Fund Data Ecosystem (CFDE) Training Center

Decoding the Data Ecosystem Institute
(DeCoDE Institute)

Part 1: Overview of CFDE_Scavenger Hunt

June 2, 2026

ORAU

CFDE TRAINING CENTER

BioData Sage

Welcome/Agenda

- CFDE Overview
- CFDE Training Center Overview
- DeCoDE Institute Overview
- CFDE Scavenger Hunt
- Q&A

CFDE Overview

NIH Common Fund Data Ecosystem (CFDE)

Sahana N. Kukke, PhD
Program Leader, NIH Common Fund

Decoding the Data Ecosystem Institute (DeCoDE Institute)
June 2, 2026

National Institutes of Health
Office of Strategic Coordination–The Common Fund

NIH CFDE Team


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David Dzamashvili, Operations (Trio)


Alexandria Burns, DOTM (Trio)

Common Fund Programs

Transformational Science and Discovery (TSD)

These **programs** are designed to establish new scientific principles models, and research resources to transform scientific knowledge and paths to discovery.

- 4D Nucleome
- A2CPS
- CryoEM
- DS-I Africa
- SMaHT
- Human Virome
- HRHR
- MoTrPAC
- NIOI
- SPARC
- NPH
- Oculomics
- SenNet
- RNomics

Re-Engineering the Research Enterprise (RRE)

These **programs** are designed to transform how we do biomedical and behavioral research, how we make the biomedical workforce as robust as possible to ensure new perspectives and ideas contribute to discovery, how we transition that research into prevention and therapies, and how those successful prevention and therapies can be shared broadly.

- CARE for Health™
- Complement-ARIE
- NBSxWGS
- Replication Initiative
- SCGE
- Transformative Research to Address Health Disparities

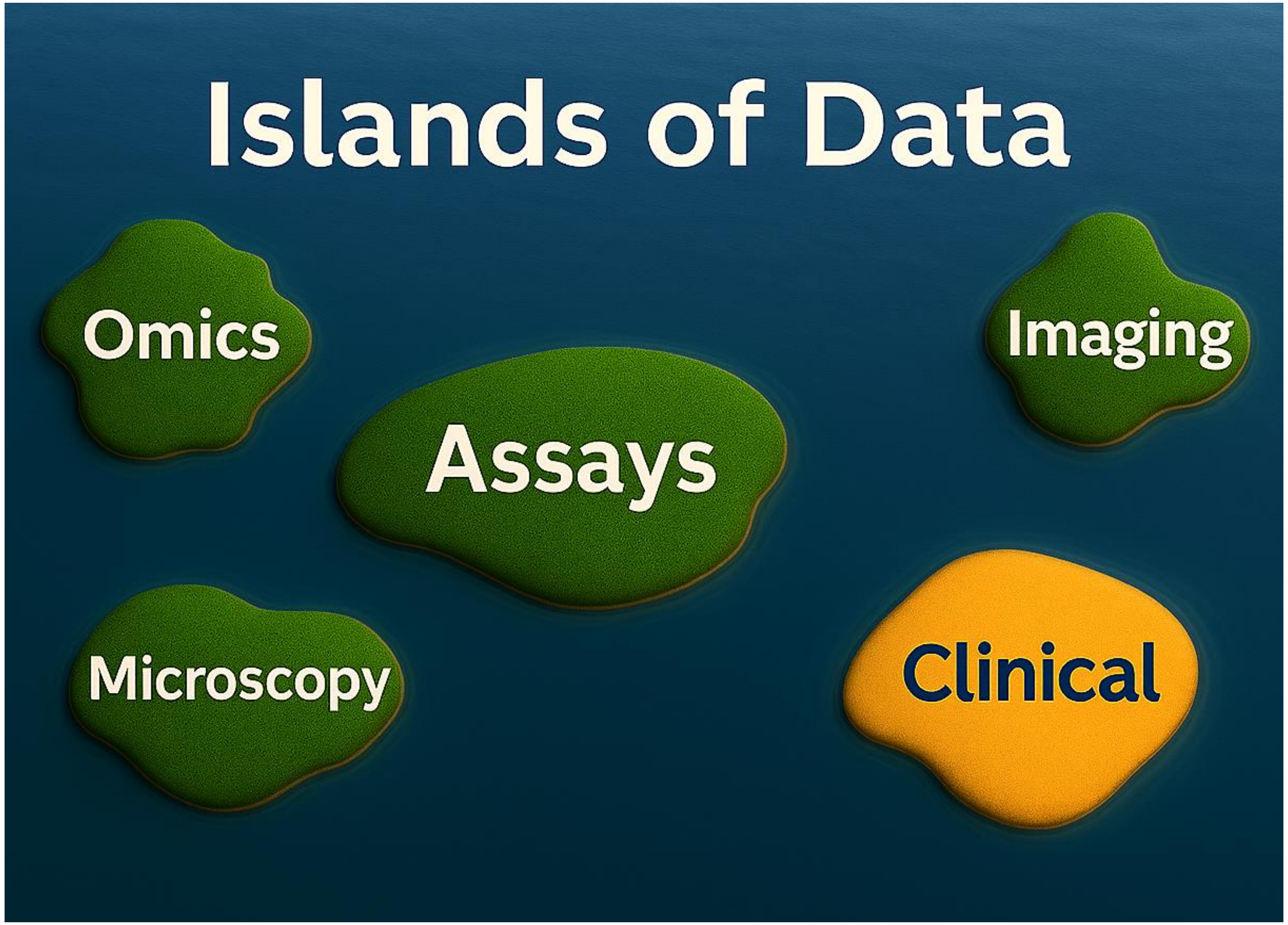
TSD programs that create catalytic data resources

4D Nucleome ◦ A2CPS ◦ Human Virome
MoTrPAC ◦ NPH ◦ SenNet ◦ SMaHT ◦ SPARC

Catalytic Data Resources (CDR)

These **programs** are designed to manage and develop data for scientific discoveries by accelerating research through data resources.

- Bridge2AI
- CFDE
- HuBMAP
- Kids First
- PRIMED-AI
- SysBio


Islands of Data
Omics
Imaging
Assays
Microscopy
Clinical

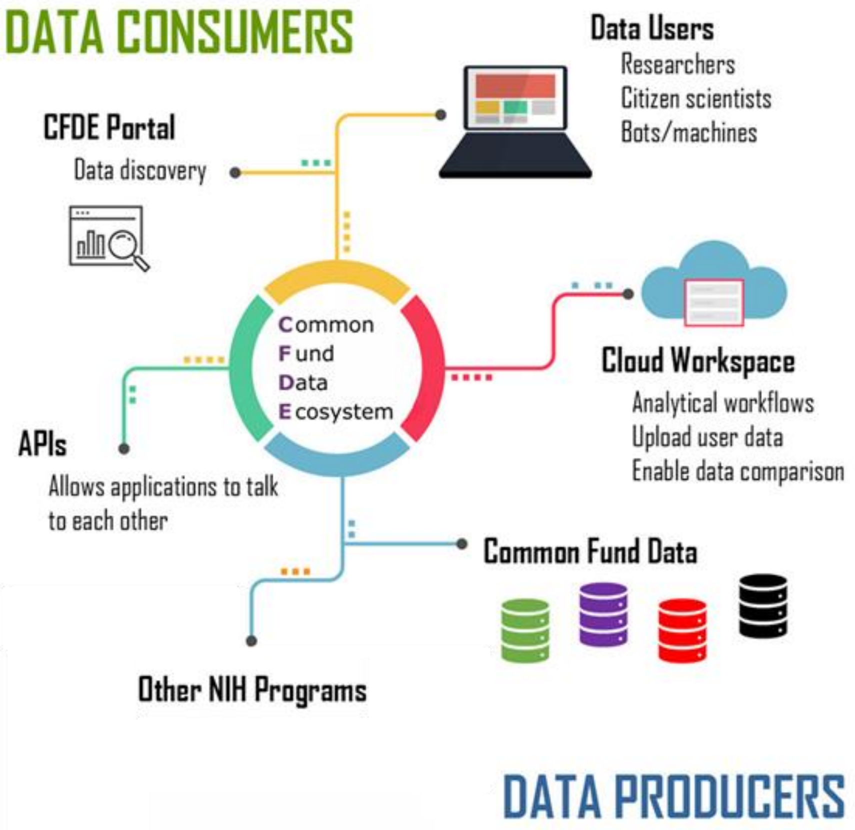

DATA SILO
DATA SILO
DATA SILO
DATA SILO

Common Fund Data Ecosystem (CFDE)

Mission: *Foster scientific discovery through the (re)use of data and resources generated by Common Fund (CF) Programs*

Aims:

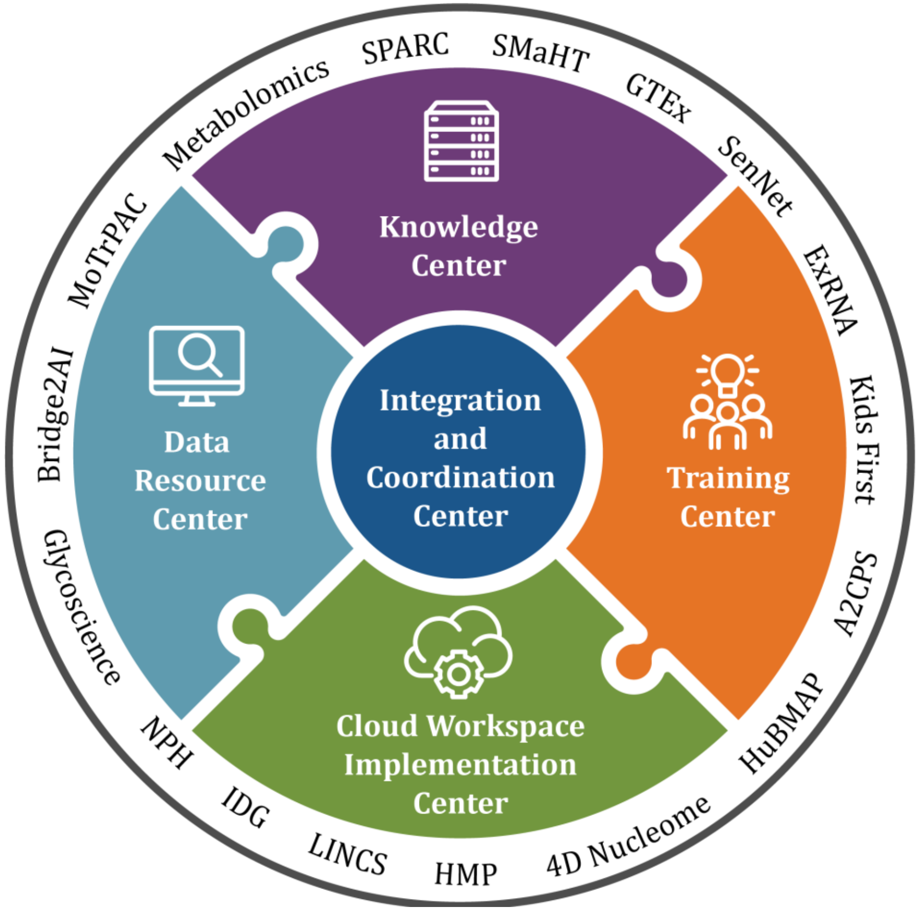
1. Enable users to **query across & use** multiple CF data sets
2. Provide **training and outreach** to bring people to CF data and train them to work in the cloud
3. **Coordinate and integrate** infrastructure and activities into a cohesive ecosystem



The chief metric of success for CFDE is discovery: If the CFDE is successful, investigators will be using Common Fund data (and products) for new discoveries and new purposes.

CFDE Approach

Core infrastructure built by five distinct but interconnected Centers



Design Pillars

1. **Build resources that are complimentary, not duplicative**
   - No data repository, federated data storage
   - Use of APIs and standards to connect with other NIH data ecosystems
2. **Community-driven ecosystem, not a platform**
   - Community-driven and maintained
   - Diversity of data and user communities means there won't be a "one-size fits all" solution
3. **Scalable and sustainable**
   - Flexible enough to evolve with the field

Discover and understand Common Fund data

CFDE WORKBENCH

https://data.cfde.cloud

CFDE DATA PORTAL

Search Common Fund Programs' Metadata and Processed Datasets.

Enter one or more keywords

Try STAT3, blood, dexamethasone

LEARN MORE   GRAPH SEARCH NEW   SEARCH →

THE KNOWLEDGE CENTER integrates data and knowledge generated by ground-breaking research programs in the NIH Common Fund Data Ecosystem.

SEARCH COMMON FUND KNOWLEDGE

Click here to begin

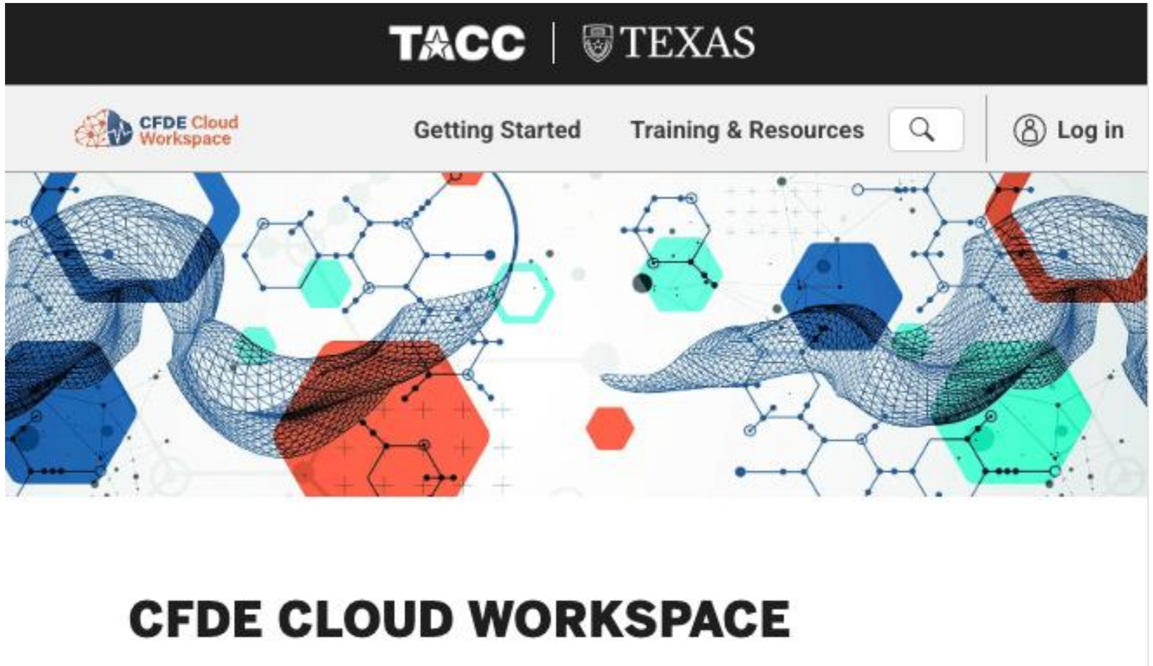
https://cfdeknowledge.org

Learn about and work with Common Fund data

Common Fund Data Ecosystem
TRAINING CENTER

Decoding the Data Ecosystem

https://orau.org/cfde-trainingcenter/



https://cfdeworkspace.org/

CFDE Training Center Overview

CFDE Training Center (TC) Overview

The TC provides training in basic and advanced computational and data analytics skills for data science learners and users interested in CFDE data, tools, and resources.

The TC aims to expand the CFDE data userbase and enhance the confidence and complexity of dataset usage through community building and engagement activities.

CFDE TC Podcast

e-Learning Dashboard

Hackathons and Workshops

DeCoDE Institute

Education & Engagement

DeCoDE Institute Learning Objectives

- Develop skills to analyze omics datasets and build workflows using CFDE tools and resources.
- Engage in collaborative, hands-on learning experiences and foster innovation through live training sessions.
- Utilize CFDE resources and e-learning modules for flexible, asynchronous learning opportunities.
- Demonstrate knowledge and skills by participating in team-based capstone projects.

Coming Soon from the Common Fund Data Ecosystem TRAINING CENTER

- **June 11: Speed Networking**
- **June 15 & 17: Part 1 & 2: Applied AI for Biologists Using Galaxy and CFDE Cloud Resources**

Submit by June 30 11:59 p.m. PT
Use data from one or more CFDE-supported programs to create a visualization that tells a compelling biomedical story. Up to **$1000 in prizes** will be awarded to winning visualizations!

Learn more and register: www.orau.org/cfde-trainingcenter

And more …

- **Monthly Podcast** – *Decoding the Data Ecosystem* is dedicated to unraveling the complexities and exploring the depths of omics research
- **e-Learning Dashboard** – Free, innovative omics training on-demand and home to the educational 'How-To Seminar Series'
- Follow us on Linked in **@cfde-trainingcenter** to stay informed of upcoming training and events!

CFDE TRAINING CENTER

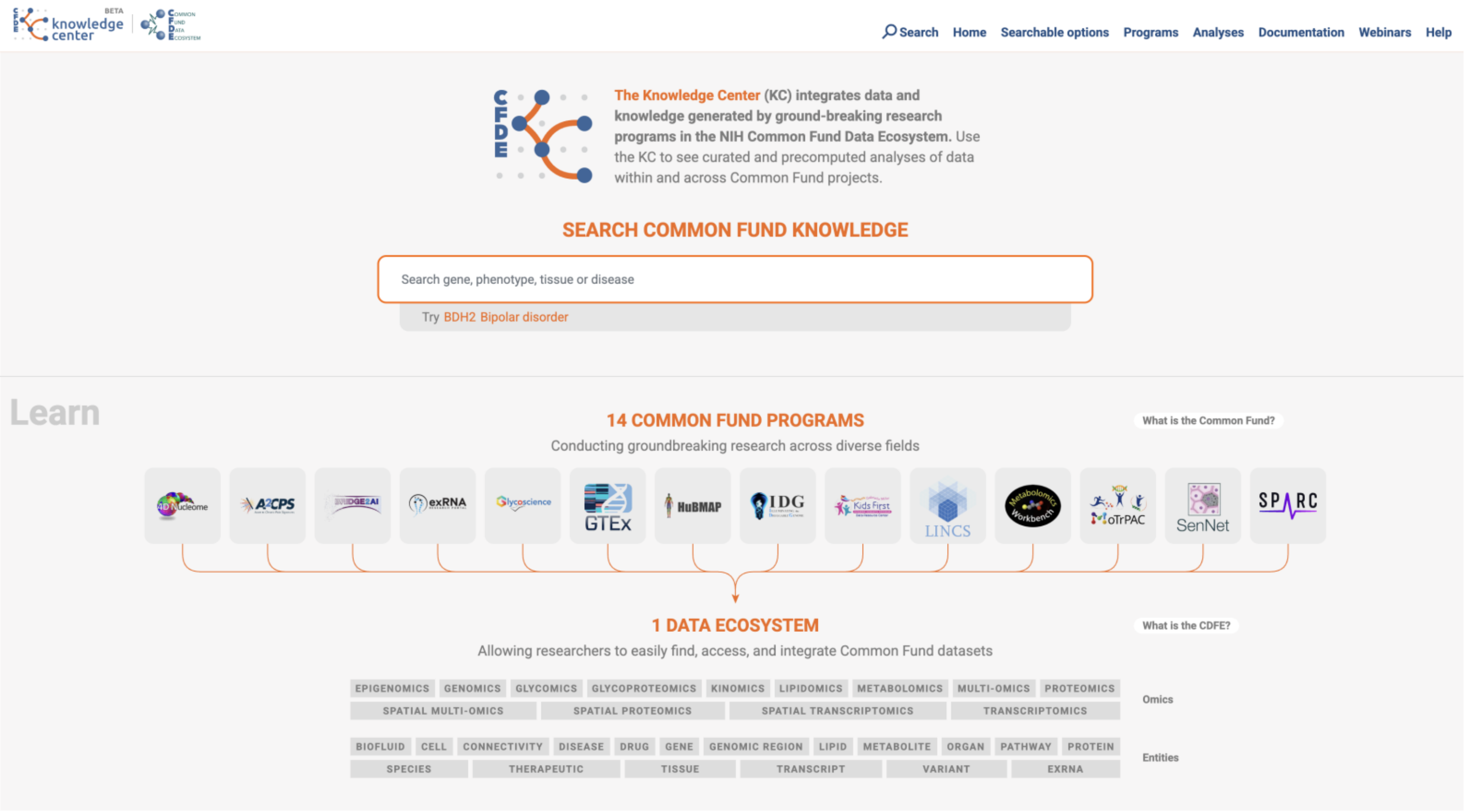
CFDE Scavenger Hunt

Now, to the activity!

- Learn to navigate the various CFDE pages:
  1. https://commonfund.nih.gov/dataecosystem
     - Information about CFDE as a whole and a snapshot of the different funded programs
  2. https://cfdeknowledge.org/r/kc_landing
     - Information about the kinds of data in each program as well as curated and precomputed analysis of data
  3. https://data.cfde.cloud/
     - CFDE workbench tools, chatbot, workflows, use cases, and more

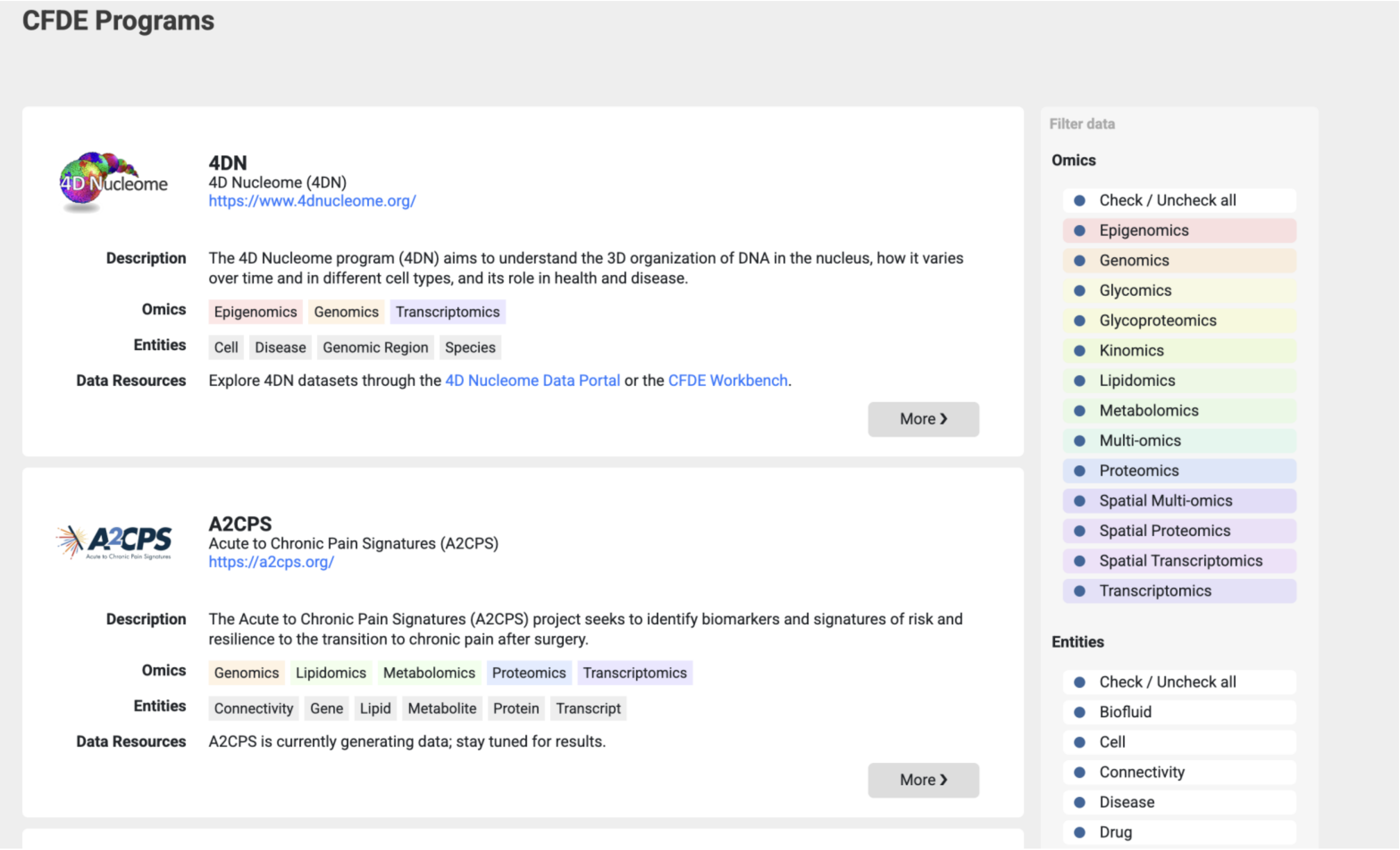
- Participate in the Scavenger Hunt:
  - Learn more about the CFDE programs and types of data found in each
  - Find and use specific CFDE tools and data
  - Identify new information for potential follow-up explorations

Who is participating in CFDE?



https://cfdeknowledge.org/r/kc_landing

What types of data are contributed?

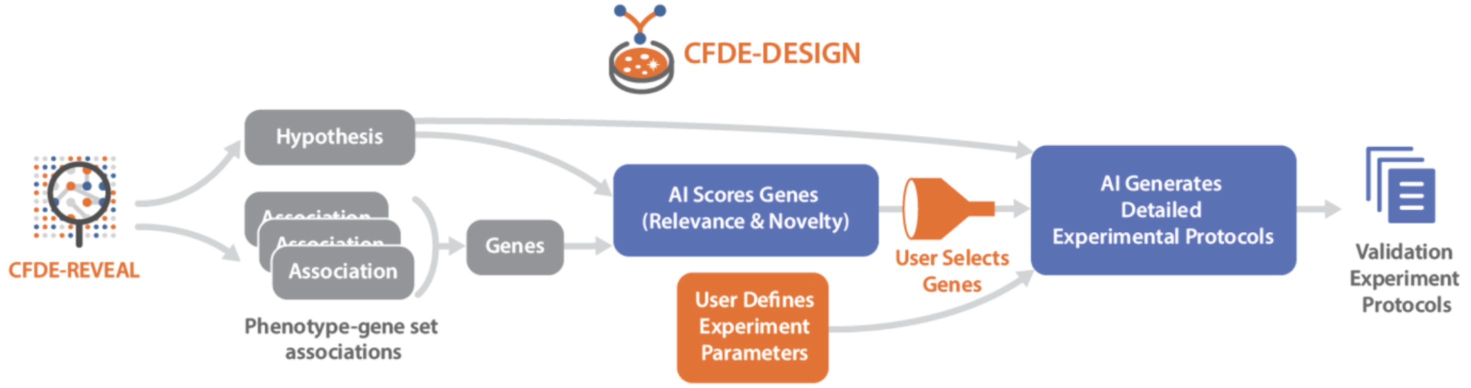


https://cfdeknowledge.org/r/kc_programs

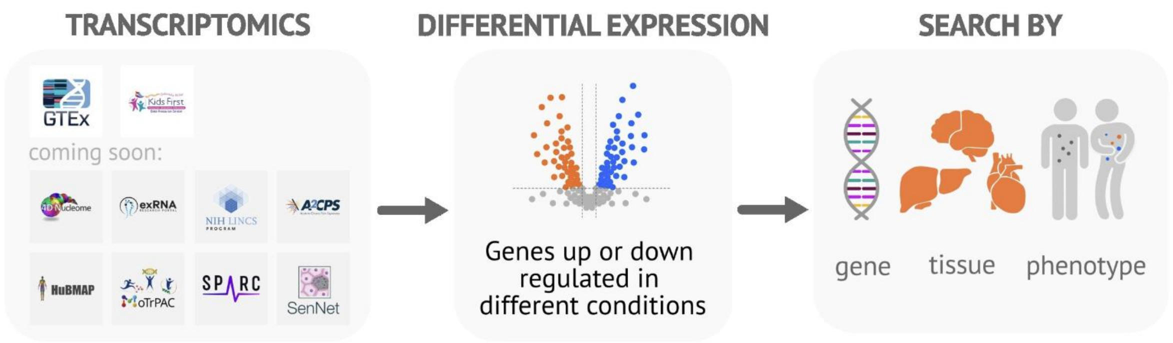
Analyses KC Labs Documentat

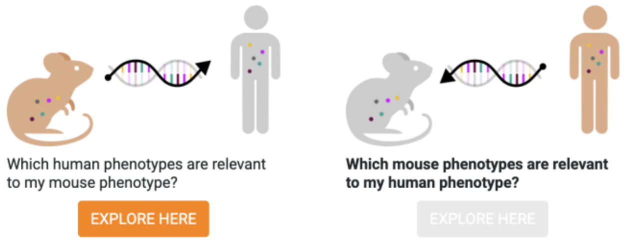
- CFDE-REVEAL
- CFDE-DESIGN
- 4DN Falcon Analysis
- Differential Expression Analysis
- Gene Set Analysis
- Mouse to Human

How can people interact with the data?



Differential Expression | All Results and Gene Sets





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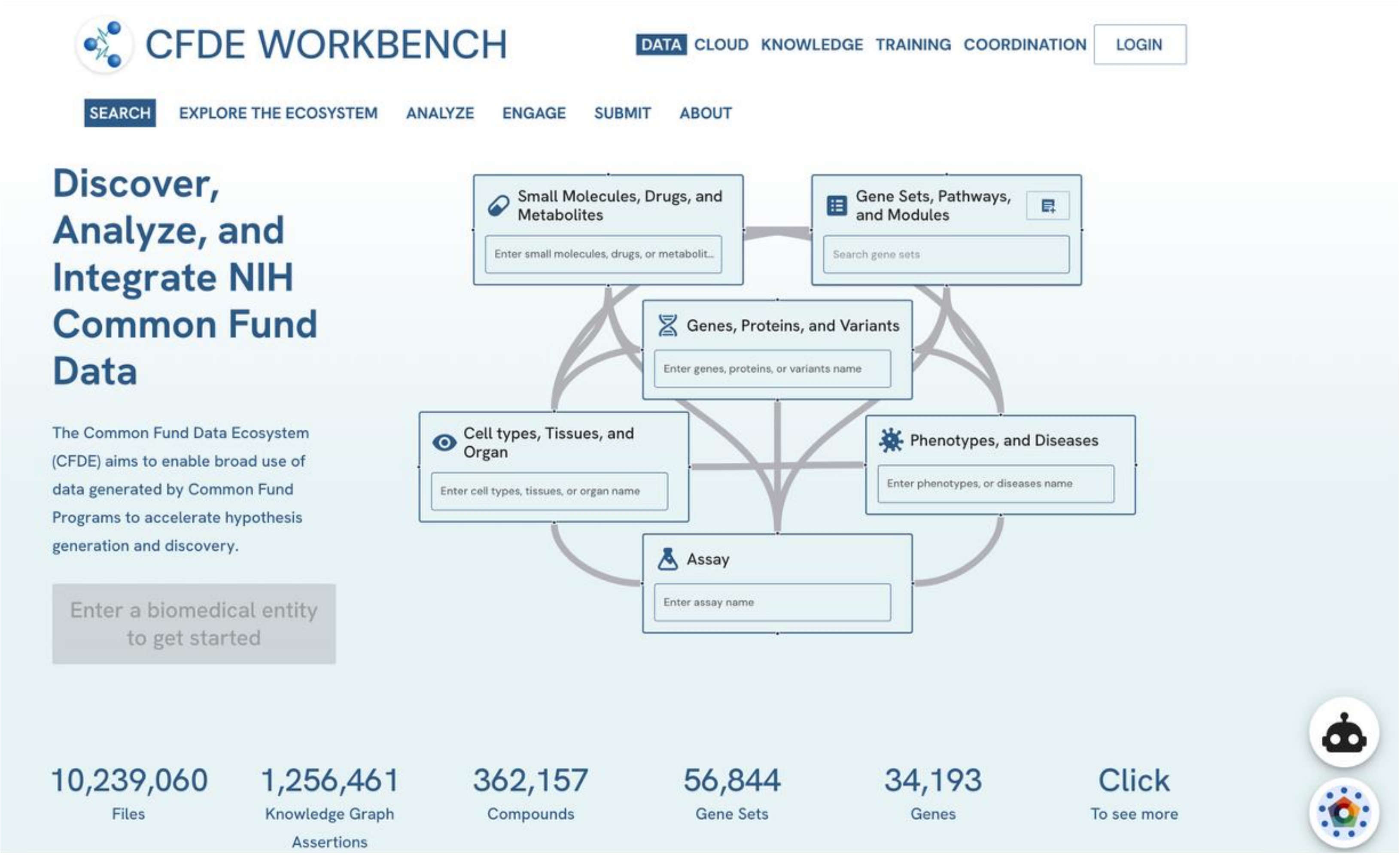

https://cfde.cloud/

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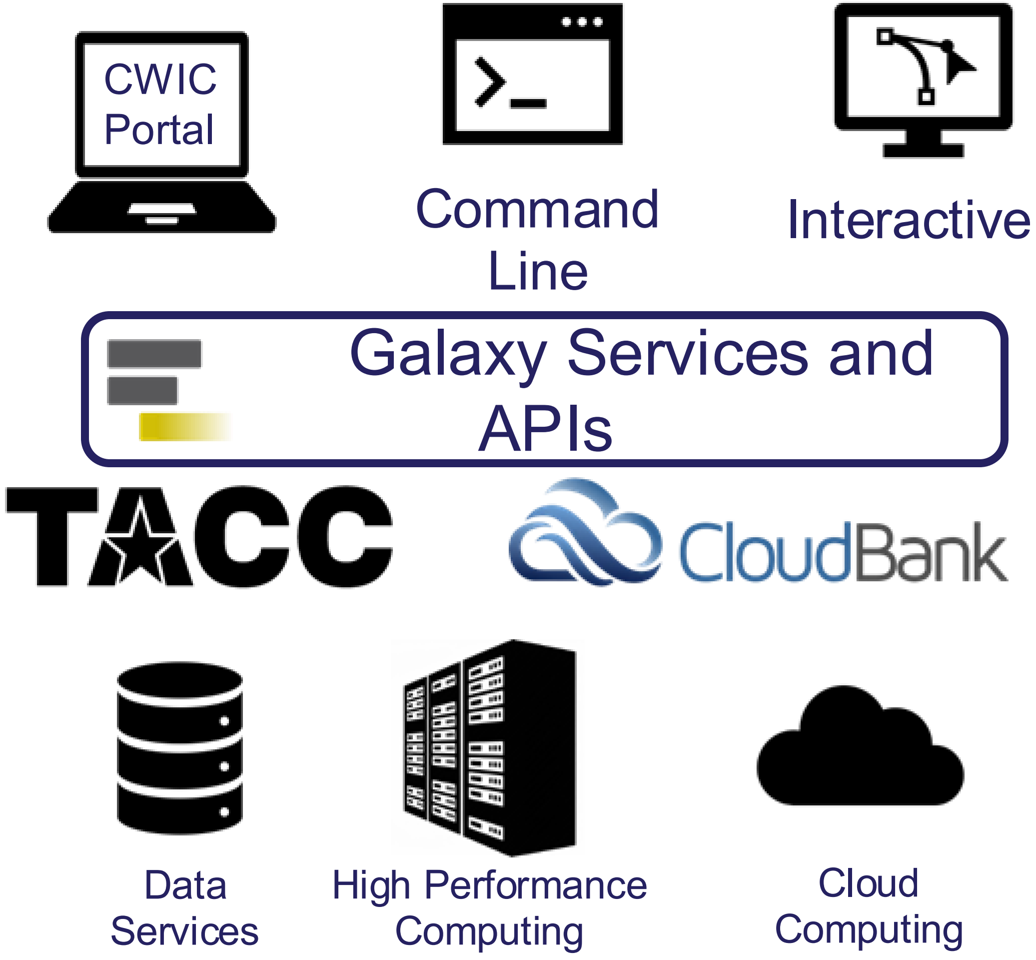
CFDE TRAINING CENTER

BioData Sage

How can people interact with the data?



https://cfdeworkspace.org/

What are researchers currently doing?

CFDE WORKBENCH

INFORMATION PORTAL DATA PORTAL LOGIN

HOME CF PROGRAMS CFDE CENTERS PARTNERSHIPS TRAINING & OUTREACH PUBLICATIONS WEBINARS WHAT'S NEW?

INFO › PUBLICATIONS

CFDE Associated and Common Fund Programs' Landmark Publications

Search publications

☑ Centers ☑ DCCs ☑ Landmark ☑ RO3s

Diwan R, Gaytan SL, Bhatt HN, Pena-Zacarias J, Nurunnabi M. 2024. **Liver fibrosis pathologies and potentials of RNA based therapeutics modalities.** *Drug Delivery And Translational Research.* 14(10):2743-2770. https://www.doi.org/10.1007/s13346-024-01551-8

PubMed DOI R03

Animals | Liver Cirrhosis | Rna | Lipid Nanoparticle | Rna Therapeutics | Mrna | Nanoparticles | Drug Delivery Systems | Sirna | Humans | Liver Fibrosis

Beaven E, Kumar R, An JM, Mendoza H, Sutradhar SC, Choi W, Narayan M, Lee YK, Nurunnabi M. 2024. **Potentials of ionic liquids to overcome physical and biological barriers.** *Advanced Drug Delivery Reviews.* 204:115157. https://www.doi.org/10.1016/j.addr.2023.115157

PubMed PMC DOI Export Citation R03

Administration, Cutaneous | Biological Barrier | Ionic Liquid | Ionic Liquids | Local Delivery | Noninvasive Delivery | Drug Delivery Systems | Systemic Delivery | Humans

Fisher JL, Wilk EJ, Oza VH, Gary SE, Howton TC, Flanary VL, Clark AD, Hjelmeland AB, Lasseigne BN. 2024. **Signature reversion of three disease-associated gene signatures prioritizes cancer drug repurposing candidates.** *Febs Open Bio.* 14(5):803-830. https://www.doi.org/10.1002/2211-5463.13796

PubMed PMC DOI Export Citation R03

Gene Expression Profiling | Cancer | Drug Repositioning | Glioblastoma | Animals | Glioblastoma | Gene Expression Regulation, Neoplastic | Gene Signature | Mice | Brain Neoplasms | Transcriptomic Signature | Xenograft Model Antitumor Assays | Transcriptome | Antineoplastic Agents | Neoplasms | Cell Line, Tumor | Drug Repurposing | Humans

Gjoni K, Pollard KS. 2024. **SuPreMo: a computational tool for streamlining in silico perturbation using sequence-based predictive models.** *Bioinformatics (oxford, England).* 40(6). https://www.doi.org/10.1093/bioinformatics/btae340

PubMed PMC DOI Export Citation R03

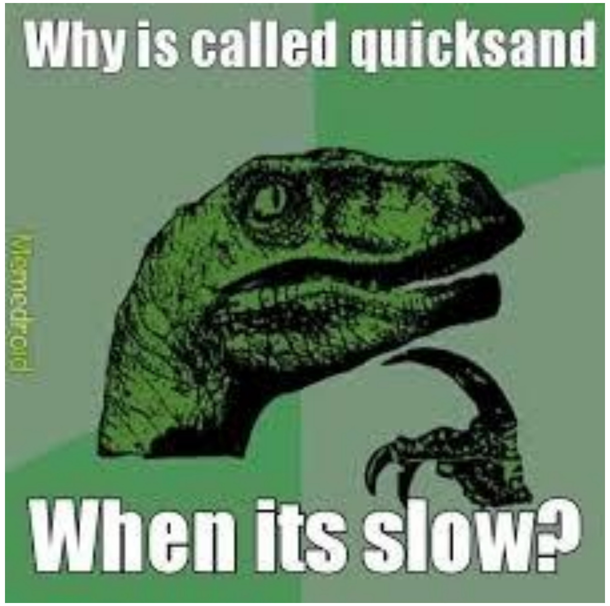
https://info.cfde.cloud/publications

Instructions

- Participants will be assigned to teams of two
- Teams will have ~30 minutes to work through the Digital Scavenger Hunt
    - The last question is open-ended: asking participants to find an interesting observation from their favorite dataset
- Answers will be reviewed together as a group with demonstrations
    - A different team will get called on to answer each question
    - There will be an opportunity to share the interesting observations
- General Q&A
- https://bit.ly/Scavenger_Hunt_template

Q & A Time!


End of Presentation
ANY QUESTIONS?


Why is called quicksand
When its slow?

Contact Us

Learn more about the Training Center:
https://orau.org/cfde-trainingcenter

Connect with us on LinkedIn

https://www.linkedin.com/company/cfde-trainingcenter

Questions? Interested in collaborating or being a podcast guest?

Email us: cfde-trainingcenter@orau.org

Please Take Our Survey

https://orausurvey.orau.org/n/DeCoDEscavengerhunt.aspx